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2 May 1980
SUPERSEDING
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23 March 1972

MILITARY SPECIFICATION

DISTILLATION UNITS, WATER, THERMOCOMPRESSION: TRAILER-MOUNTED, 150 GALLONS PER HOUR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers military design trailer-mounted, gasoline- and diesel-engine-driven, thermocompression, distillation units.

1.2 Classification. The distillation units shall be of the following types, as specified (see 6.2):

Type I - Distillation unit, trailer-mounted, gasoline-engine-driven, 150 gallons per hour (gph).
Type II - Distillation unit, trailer-mounted, diesel-engine-driven, 150 gph.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 4620

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

FF-N-105	- Nails, Brads, Staples and Spikes: Wire, Cut and Wrought.
NN-P-530	- Plywood, Flat Panel.
QQ-S-781	- Strapping, Steel, and Seals.
VV-F-800	- Fuel Oil, Diesel.
VV-G-1690	- Gasoline, Automotive, Low Lead or Unleaded.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-T-60	- Tape: Packaging, Waterproof.

MILITARY

MIL-V-173	- Varnish, Moisture-and-Fungus Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-D-3209	- Distillation Units, Water, Thermocompression-Type, Packaging of.
MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Guns, Hand, High-Pressure, Lever-Operated; Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity).
MIL-G-5572	- Gasoline, Aviation, Grades 80/87, 100/130, 115/145.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-C-11029	- Citric Acid.
MIL-S-12165	- Strainers, Suction, Fire Hose; and Strainers, Suction, Hose.
MIL-F-16884	- Fuel Oil, Diesel, Marine.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-C-52437	- Chassis, Trailer, Single- and Multi-Axle.

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STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-130	- Identification Marking for US Military Property.
MIL-STD-461	- Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-781	- Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1400	- Engines, Gasoline and Desiel, Methods of Test.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.
MIL-STD-1474	- Noise Limits for Army Material.
MS35644	- Filler Neck: Fuel Tank-Military Vehicles.
MS35645	- Cap, Fuel Tank: Tactical Vehicle, Fording.
MS35844	- Plug, Machine Thread, Magnetic Drain.
MS49006	- Plugs, Pipe, Magnetic-Headless; Aluminum, Iron, or Steel.
MS53063	- Indicator, Air Cleaner, Intake; Restriction Mechanical Type.

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DRAWINGS

ME

TA13211E4485	- Distillation Unit, Water, Thermo-compression; Trailer-Mounted, Gasoline-Engine-Driven; 150 GPM, Type I.
TA13215E7032	- Distillation Unit, Water, Thermo-compression; Trailer-Mounted, Diesel-Engine-Driven; 150 GPH, Type II.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN PUBLIC HEALTH ASSOCIATION (APHA)

Standard Methods for Examination of Water and Wastewater.

(Application for copies should be addressed to the American Public Health Association, Inc., 1015 Eighteenth Street, NW, Washington, DC 20036.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX - Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

B3.0 - Standard Qualification Procedure.

(Application for copies should be addressed to the American Welding Society, 2501 Northwest Seventh Street, Miami, FL 33125.)

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HYDRAULIC INSTITUTE (HI)

Centrifugal Pumps, Rating Standards and Test Code Standards of the Hydraulic Institute.

(Application for copies should be addressed to the Hydraulic Institute, 1230 Keith Building, Cleveland, OH 44115.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemically Reactive Solvents.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 South San Pedro Street, Los Angeles, CA 90013.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Title 40, Code of Federal Regulations, Part 141, National Interim Primary Drinking Water Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The distillation units shall be in accordance with Top Assembly TA13211E4485 for Type I and Top Assembly TA13215E7032 for Type II and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting

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officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the distillation unit. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one distillation unit for examination and test within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce distillation units that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examination and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more distillation units for inspection as specified in 4.5.

3.4 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.13). The material selected shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the procuring activity to the appropriate medical service who will act as an advisor to the procuring activity (see 6.4).

3.5 Threads. All threads shall be in accordance with FED-STD-H28.

3.6 Environmental.

3.6.1 Operating temperature. The distillation unit shall perform as specified herein in any ambient temperature from plus 110° F to minus 25° F (plus 43° C to minus 32° C).

3.6.2 Storage temperature. The distillation unit shall not be damaged by storage at ambient temperatures from plus 160° F to minus 30° F (plus 71° C to minus 34° C).

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3.7 Safety. All shafts or rotating or moving parts of the distillation unit shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of the equipment, including exhaust pipes exposed to contact by personnel or which create a fire hazard, shall be fully guarded or insulated. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto the engine, exhaust, or electrical equipment. Exhaust or discharges from the equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard.

3.8 Human factors. The distillation unit shall conform to the applicable human factors engineering criteria as specified in MIL-STD-1472.

3.8.1 Noise limits. The noise of the distillation unit (excluding horns and warning alarms) shall not exceed the limits specified in MIL-STD-1474, Category D for steady-state noise when tested in accordance with 4.7.2.6.1. The precautions of MIL-STD-1474 (see 6.10) shall be provided if the sound level exceeds 85 decibels audible range (dB(A)) when tested in accordance with the distillation unit. Hazard signs, when required, shall state "CAUTION, HEARING PROTECTION REQUIRED". Plates shall be in accordance with MIL-P-514, Type III, Composition C of Type I, Grade A, Class 1 material.

3.9 Reliability. The specified mean-time-between-failure (MTBF) shall be 240 hours when the distillation unit is tested as specified in 4.7.2.7.

3.10 Maintainability.

3.10.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts, as specified (see 6.8 and 6.9), shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.7.2.8.

3.10.2 Maintenance ratio. The distillation unit shall have a maintenance ratio of not more than 0.11. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

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3.11 Performance. The distillation unit shall meet the following requirements:

(a) The distillation unit shall operate utilizing seawater with not more than one 2-hour shutdown for preventive maintenance and lubrication during any 24-hour period.
(b) Initial distillate shall be produced within 1 hour and 15 minutes of starting at any feed-water temperature from plus 32° F to plus 90° F (0° C to plus 32° C), without the use of auxiliary devices.
(c) The distillate flow rate shall be not less than 150 gph with the water-to-fuel ratio on a weight per hour basis not less than 150 to 1.
(d) The temperature of the distillate shall be not more than plus 15° F (plus 9° C) above the feed-to-water temperature.
(e) The total chloride content of the distillate shall not exceed 20 milligrams per liter (mg/L) when reported as chlorides, and the brine waste concentration shall be not more than 3.0 times the salinity of the feed water.
(f) The brine waste flow rate shall be not less than one-third of the feed-water flow rate.

3.12 Engine.

3.12.1 Gasoline. The gasoline engine, including all systems, components, accessories, and auxiliaries, shall be a heavy-duty 4-cycle liquid cooled standard commercial-industrial type capable of operating on gasoline conforming to VV-G-1690 and MIL-G-3056 and lubricating oil as specified herein. If the normal operating oil pressure of the engine is 15 pounds per square inch (psi) or above, a full-flow oil filter shall be furnished. A dry-type air cleaner shall be furnished. The temperature of the coolant in the radiator top tank shall not exceed plus 210° F (plus 99° C) when the engine is operated under any condition specified herein. The engine shall have both manual and electrical starting systems. The electrical battery starting and charging system shall be 12-volt (V) negative ground. The engine shall be furnished with a housing and shall be the same engine that has been used for at least 1 year in the manufacturer's standard commercial distillation unit of the same size and capacity.

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3.12.2 Diesel. The diesel engine, including all systems, components, accessories, and auxiliaries, shall be a heavy-duty liquid cooled standard commercial-industrial type capable of operating on diesel fuel conforming to VV-F-800 and lubricating oil as specified herein. If the normal operating oil pressure of the engine is 15 psi or above, a full-flow oil filter shall be furnished. A dry-type air cleaner shall be furnished. The temperature of the oil in the oil sump shall not exceed plus 250° F (plus 121° C) and the temperature of the coolant in the radiator top tank shall not exceed plus 210° F (plus 99° C) when the engine is operated under any conditions specified herein. The engine shall be equipped with a 12-V negative ground electrical starting and charging system. The engine shall be equipped with a housing. The engine shall be the same engine which has been used for at least 1 year in the manufacturer's standard commercial distillation unit of the same size and capacity.

3.12.3 Power and speed rating. The power and speed rating of the gasoline or diesel engine shall be such that operation of the distillation unit under any of the operating conditions specified herein will not require horsepower in excess of the continuous rating of the engine as previously published by the engine manufacturer, or 85 percent of the engine manufacturer's previously published maximum horsepower rating at the applied speed, whichever is less.

3.12.4 Filling, draining, and checking provisions. The engine crankcase shall be equipped with a dipstick to determine the fluid levels. The drain outlet shall be fitted with a magnetic drainplug having either tapered pipe threads or shoulder and gasket, size conforming to MS35844 or MS49006. In lieu of a magnetic drainplug, a separate magnetic device may be fastened inside the enclosure at or near the bottom and shall be removable for cleaning without removal of other parts or components. Each drainplug or valve shall be located so that removal of the plug or valve will result in complete drainage of the fluid from the enclosure. When the distillation unit is in a level position, drainage of fluids shall be to the ground without draining on or over any part of the distillation unit. Integral tubes, hose, or troughs may be used to convey the fluids from the drain outlet. Accessibility of the fill opening, the fluid level checking device, and the drainplug or valve shall be provided without removal or adjustment of accessories and parts except plates equipped with hand-operable quick-disconnect fasteners or capscrews in applications where quick-disconnect fasteners are subject to damage. The drain outlets for the engine cooling system shall provide complete drainage of the system. The radiator drain shall be such that coolant will not drain onto any part of the distillation unit. An extension hose shall be permitted.

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3.12.5 Instrumentation. An instrument panel shall be furnished and shall be equipped with the following instruments:

(a) Lubricating oil pressure gage.
(b) Tachometer.
(c) Ammeter, battery charging.
(d) Hourmeter.
(e) Cooling liquid temperature indicator.
(f) Air cleaner restriction indicator conforming to MS53063.
(g) Fuel pressure gage (diesel).

The tachometer and hourmeter may be incorporated into one instrument provided the hourmeter corresponds to the clock time at the normal operating speed of the distillation unit.

3.12.6 Control device system. The engines shall be provided with a high cooling liquid temperature device for stopping the engine if the engine reaches an overheated condition. Engines shall be provided with a low lubricating oil pressure control device for stopping the engine when a low lubricating oil pressure exists. Each safety device shall have individual signal lights and manual reset (if reset is required) mounted on the instrument panel. The signals shall indicate which condition has caused the shutdown and shall be marked as to their function. All control devices shall be designed to actuate at the engine manufacturer's recommended temperatures and pressures.

3.12.7 Manual engine controls. Manual engine controls for gasoline engines shall include an ignition switch to activate the entire electrical circuit, hand choke, and throttle. Manual engine controls for diesel engines shall include a switch to activate the engine electrical circuits, including fuel solenoid valve, safety control devices, and starter and battery charging circuits. A waterproof push-button starter switch shall be furnished. The throttle shall be of the micrometer type and shall permit operation of the engine at any selected governed speed from idling to maximum governed speed. The throttle shall be provided with a convenient and positive means for locking in any position.

3.12.8 Fuel tank. A fuel tank shall be furnished for both gasoline and diesel engines. The capacity of the tank shall be sufficient for not less than 4 hours of continuous operation at the rated load. The fuel tank shall be equipped with a cap conforming to MS35645, a strainer, and a neck conforming to MS35644. The tank shall be provided with a

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hand-operated fuel shutoff valve attached directly to the tank and a mechanically actuated fuel gage mounted on the tank. The fuel tank outlet shall be not less than 0.50 inch above the bottom of the tank. The tank outlet or the fuel line shall be provided with a 3-way selector valve and a 0.50-inch 20 SAE flared fitting for connection with an auxiliary fuel supply. A corrosion-resisting circular metal face plate shall be mounted on the fuel selector valve and shall be marked "OFF", "UNIT TANK", and "AUXILIARY".

3.12.9 Battery-disconnect switch. The distillation unit shall be provided with a battery-disconnect switch to permit disconnecting the battery in the event of an emergency. The switch shall be installed in a convenient position on the instrument panel and shall withstand the full cranking motor load.

3.13 Mobility. The distillation unit, mounted on the specified trailer, shall withstand without evidence of damage, permanent deformation, and leakage, the following:

(a) Travel over smooth, hard-surface primary highways at variable speeds up to 50 miles per hour (mph).
(b) Travel over unimproved secondary roads at variable speeds up to 25 mph.
(c) Travel over open, rolling, and hilly cross-country terrain at variable speeds up to 10 mph.

3.14 Electromagnetic interference. The electromagnetic interference emmission and susceptability characteristics of the distillation unit shall conform to MIL-STD-461, Class III C.

3.15 Lubrication. Except the engine, all surfaces requiring lubrication shall be provided with a means for lubricating.

3.15.1 Lubricants. The distillation unit shall operate as specified herein when lubricated with military lubricants (see 6.11).

3.15.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, Types I, II, or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility of fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

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3.15.3 Pressure-release. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease. seals or other parts.

3.15.4 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

(a) MIL-G-23827 - For ball, roller, and needle bearings, gears, and on sliding and rolling surfaces of such equipment as instruments, cameras, electronic gear and mechanical control systems; for use on gears, actuator screws and other equipment requiring a lubricant with high load-carrying capacity over a temperature range of minus 100° F to plus 250° F (minus 73° C to plus 121° C).
(b) MIL-G-10924 - For lubrication and surface corrosion protection of automotive and artillery equipment operated over the temperature range of minus 65° F to plus 225° F (minus 54° C to plus 107° C). Also used in other applications within this temperature range where grease with oxidation-resistant and corrosion-preventive properties is desirable. Not for use on machinery which comes in contact with foods or potable water.

The distillation units shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not always compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.16 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.16.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose function will be adversely affected by the varnish coating shall not be treated.

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When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.16.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish preclude the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 102, Southern California Air Pollution Control District.

3.17 Identification marking. The distillation unit shall be identified in accordance with MIL-STD-130. The marking shall be applied to the distillation unit on plates conforming to MIL-P-514, Type I, Style 1, Composition C of Type I, Grade A, Class 1 material. The plates shall be attached to the distillation unit with adhesive specified on applicable drawings.

3.18 Shipping data plates. Two shipping data plates shall be furnished. One shipping plate shall show in silhouette the location of lifting and tiedown attachments when unmounted. The other shipping plate shall indicate the location of lifting and tiedown attachments when mounted on the trailer. The shipping data plates shall conform to MIL-P-514, Type I, Style 1, Composition C of Type I, Grade A, Class 1 material. The plates shall be attached to the distillation unit unmounted and mounted, with adhesive specified on applicable drawings.

3.19 Designation plates. Each valve shall be designated in accordance with Figure 1A or 1B as applicable. Designation plates shall be in accordance with MIL-P-514, Type III, Composition C of Type II, Grade A, Class 1 material, 1 inch by 2 inches, white printing on a black background on both sides. Plates shall be attached to the applicable valves with 0.061-inch diameter, corrosion-resistant wire.

3.20 Instruction plates. Each distillation unit shall be provided with operating instruction plates and flow diagrams (Figure 1A and 1B as applicable), including warning, caution, and safety data, describing any special or important procedures to be followed in assembling, operating, or servicing the distillation unit. Plates shall conform

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to MIL-P-514, Type III, Composition C of Type I, Grade A, Class 1 material, size as applicable, and shall be attached to or near the operating area with rivets, screws, or with adhesive specified on applicable drawing. Where attachment by rivets, screws, or adhesive is impractical, plates shall be attached with 0.061-inch diameter, corrosion-resistant wire. Marking of the caution plate shall be white printing on a red background, as specified in MIL-P-514.

3.21 Government-furnished property. When specified (see 6.2), the following property will be furnished by the Government (see 6.6):

Item No.	Description	Identification	Quantity
1.	Chassis, Trailer Trailing Arm, 2-1/2 Ton	MIL-C-52437	1

3.22 Government-loaned property. Unless otherwise specified (see 6.2), the following property will be loaned by the Government (see 6.7):

Item No.	Description	Quantity
1.	M44 or M45 Series Truck, 6 x 6, 2-1/2 Ton	1

3.23 Transportability.

3.23.1 Rail. Unless otherwise specified (see 6.2), the unpackaged distillation unit shall withstand rail humping without damage or permanent deformation to any part or component when tested as specified in 4.7.2.17. After testing, the distillation unit shall operate as specified herein for a period of not less than 1 hour without malfunction or permanent deformation of any part of the distillation unit.

3.24 Workmanship. All parts, components, and assemblies of the distillation unit including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. All edges shall be rounded or chamfered. The unit shall be free from such defects as misalined components, incomplete welds, rust, extraneous material, cracks, leaks, and other defects that could impair the operation and serviceability of the unit.

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3.24.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.24.2 Castings. Castings shall be sound and free from patching, misplaced coring, warping, porosity, or other defects which might render the casting unsound for its intended use.

3.24.3 Welding procedures and qualifications.

3.24.3.1 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the distillation unit to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.1875 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the distillation unit shall be at the option of the contractor.

3.24.3.2 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

B3.0 - Standard Qualification Procedure of the American Welding Society.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

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Contractors who make only horizontal welds neet not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS or ASME codes.

3.24.3.3 Aluminum and steel weldments. The surfaces and edges of parts to be welded including layers of weld materials when multiple layers are required, shall be free from oxide scale, paint, lubricant, and other foreign matter. Welds shall transmit stress to the heat-affected zone without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading. All welded parts shall be free from cracks, porosities, and other imperfections that may reduce the effectiveness of the part. Welds involving corrosion-resistant steel shall be passivated and annealed as specified in applicable drawing.

3.24.3.3.1 Aluminum welding. All aluminum welds shall be made at a uniform rate of speed. Complete and uniform penetration and fusion of the metals shall be obtained on all welds without metallic and nonmetallic inclusions. Unless otherwise specified (see 6.2), finished ALCLAD aluminum welds shall not be ground, filed, chipped, or hammered, but may be wire brushed. Preheating for welding is permissible provided the temperature does not exceed plus 300° F (plus 149° C) for a total time not to exceed 30 minutes. No process requiring the use of welding flux shall be used. Work shall be positioned and jigged for flat welding whenever practicable.

3.24.3.4 Weldment specimens. The contractor shall submit steel and aluminum specimens representing each typical weld joint utilized in fabrication of the distillation unit. These specimens shall be furnished at the same time as the first article distillation unit. Weldment specimens shall be submitted by each welder performing weldments on the distillation unit with each type of welding equipment used. Acceptable specimens shall represent the minimum acceptable weld quality and shall be retained as inspection standard for evaluation of production weldments.

3.24.3.5 Welding procedure. Prior to any welding, the contractor shall prepare and submit to the contracting officer for approval of written welding procedure defining process and control. This procedure

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shall include, but not be limited to, those factors noted in Table I. Welding procedure changes affecting any of the factors in Table I must be approved by the contracting officer prior to adoption of the change.

TABLE I. Factors to be included in joint welding procedures.

Item number	Factors
1.	Welding process and techniques including shielding medium.
2.	Welding process operation.
3.	Type of current, ac or dc, and polarity of dc if used.
4.	Shielding gas composition.
5.	Joint design.
6.	Method of joint preparation.
7.	Preweld and interpass cleaning.
8.	Preheat or ambient temperature range.
9.	Composition of filler metal.
10.	Filler metal size for all passes.
11.	Number and sequence of passes.
12.	Welding position.
13.	Repair method to be used in removing defects.
14.	Method of inspection used to insure proper repair.
15.	Method for qualification of welders.
16.	Method for determining capability of automatic equipment and equipment operators.
17.	Workmanship specimens.
18.	Method of inspection of finished welds.

3.24.3.6 Rewelding defects. Except ALCLAD material, portions of welded joints that have been rejected on inspection because of defects cited in 3.24.3.3 or not conforming to weldment specimens (see 3.24.3.4) may be repaired as specified on applicable drawing only by rewelding. The defective weld shall be removed by chipping or machining. With the exception of steel, flame cutting shall not be used. Before rewelding, the joint shall be inspected to assure that all of the defective weld has been removed and that the joint is accessible so that the welding operator can obtain full penetration and fusion through the joint. Repairs in the form of patching of ALCLAD aluminum shall not be performed.

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3.24.3.7 Jigs and fixtures. Components except minor parts shall be assembled in jigs or frames and joined while held in position. Jigs or frames shall be designed to minimize distortion of the parts being welded.

3.24.4 Aluminum fabrication. Care shall be used in handling aluminum during fabrication to avoid scratching or damage to ALCLAD surfaces and edges. Sharp instruments shall not be used for marking bend lines. Aluminum shall not be flame cut, but shall be cut by routing, shearing, or sawing. Holes for aluminum shall be drilled, and all burrs and rough edges shall be removed. Punch prick marks shall not exceed 5 percent of the material thickness, nor be employed on ALCLAD surfaces.

3.24.5 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

3.24.6 Diagrams, charts, and plates. Diagrams, charts, and plates attached to components with adhesive shall be applied to give a surface free of wrinkles, bubbles, or other defects that may cause the diagrams, charts, and plates to become loose or damaged.

3.24.7 Cleaning. The distillation unit and components shall be thoroughly cleaned of all cutting oil, grease, residue of tests, dirt, metal scrap, and other contaminants. Cleaning shall be accomplished in a manner that will not leave a residue or otherwise make the distillation unit and components unsuitable for its intended use. The unit shall be completely drained and dried before preparation for delivery.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the preproduction model shall be cause for disassembly, in the presence of a Government representative, of the preproduction model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the preproduction model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor. Government-furnished and Government-loaned property shall be excluded from the requirements of this paragraph.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the supplementary quality assurance provisions (SQAP) shown on drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.5.1 and 6.12).

4.1.4 End Item Final Inspection Requirement (EIFIR). Each distillation unit shall be inspected in accordance with EIFIR-EF 13211E4485 for Type I and EIFIR-EF 13215E7032 for Type II. One copy of a completed EIFIR shall be shipped with each inspected distillation unit (see 6.5.2).

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Individual component inspection (see 4.3).
(b) Preproduction inspection (see 4.4).
(c) Initial production inspection (see 4.5).
(d) Quality conformance inspection (see 4.6).
(e) Inspection comparison (see 4.8).
(f) Inspection of packaging (see 4.9).

4.3 Individual component inspection. Inspection shall be performed prior to assembling and before insulation is applied to each component as specified herein.

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4.3.1 Examination. Individual components shall be examined for defects marked "X" in Column 1 of Table II.

4.3.2 Tests. Upon successful completion of the examination specified in 4.3.1 the individual components shall be subjected to the tests marked "X" in Column 1 of Table III.

4.3.3 Inspection failure. Evidence that any component does not conform to the applicable drawing, or failure of any examination or test, shall be cause for performing the inspection specified in 4.1.2.

4.4 Preproduction inspection.

4.4.1 Examination. The preproduction distillation unit shall be examined for defects marked "X" in Column 2 of Table II. Presence of one or more defects shall be cause for performing the disassembly inspection specified in 4.1.2.

4.4.2 Tests. The preproduction distillation unit shall be subjected to the tests marked "X" in Column 2 of Table III. Failure of any test shall be cause for performing the disassembly inspection specified in 4.1.2.

4.5 Initial production inspection. When specified (see 3.3), one or more initial production distillation units will be selected at random by the Government from the distillation units being produced by production tooling and will be examined for the defects marked "X" in Column 3 of Table II and subjected to the tests marked "X" in Column 3 of Table III to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production distillation unit shall not exclude the remaining distillation units from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

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4.5.1 Inspection failure. Failure of an initial production distillation unit to meet any requirement specified herein during and as a result of the examination and tests specified in 4.5 shall be cause for rejection of the initial production distillation unit and shall be cause for refusal by the Government to continue acceptance of production distillation units until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on distillation units previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all distillation units accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.6 Quality conformance inspection.

4.6.1 Examination. Each production distillation unit shall be examined for defects marked "X" in Column 4 of Table II. Presence of one or more defects shall be cause for rejection.

4.6.2 Tests. The distillation unit shall be subjected to the tests marked "X" in Column 4 of Table III. Failure of any test shall be cause for rejection.

4.7 Inspection schedule.

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4.7.1 Examination.

TABLE II. Examination schedule.

Individual component	Preproduction	Initial production	Quality conformance	Defects	Requirement paragraph
1	2	3	4	5	6
X	X	-	X	101. Any part or component not in accordance with the SQAP requirements as shown on the drawings.	3.1, 3.1.1
X	X	-	X	102. Dimensions other than those identified on the SQAPs not as specified.	3.1, 3.1.1
X	X	-	X	103. Material not as specified.	3.4
X	X	-	X	104. Engine assembly incorrect or not complete.	3.12
X	X	X	X	105. Filling, draining, and checking provisions not as specified.	3.12.4
X	X	-	X	106. Lubricants not as specified.	3.15.1
X	X	X	X	107. Location and type of lubrication fittings not as specified.	3.15.2
X	X	X	X	108. Pressure release device not as specified.	3.15.3
X	X	-	X	109. Fungus and moisture resistance not as specified.	3.16

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TABLE II. Examination schedule. - Continued

Individual component	Preproduction	Initial production	Quality conformance	Defects	Requirement paragraph
1	2	3	4	5	6
-	X	X	X	110. Identification marking, shipping data, designation, or inspection plates missing, incomplete, or illegible.	3.17, 3.18, 3.19, 3.20
-	X	X	X	111. Information on identification, shipping data, designation, and instruction plates incorrect or illegible.	3.17, 3.18, 3.19, 3.20
X	X	-	X	112. Workmanship not as specified.	3.24
X	X	-	X	113. Cleaning not as specified.	3.24.7

4.7.2 Tests.

4.7.2.1 Test conditions. Prior to test, the distillation unit shall be lubricated with oils and greases specified in 3.15. Oils shall be those designated for use in the ambient temperature at the place of test.

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4.7.2.2 Test schedule.

TABLE III. Test schedule.

Individual component	Preproduction	Initial production	Quality conformance	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6	7
-	-	X	-	Environmental	4.7.2.3	3.6
-	-	X	-	Safety	4.7.2.4	3.7
-	-	X	-	Human factors	4.7.2.5	3.8
-	-	X	-	Noise level	4.7.2.6	3.8.1
-	-	X	-	Reliability (MTBF)	4.7.2.7	3.9
-	-	X	-	Maintainability	4.7.2.8	3.10
X	-	-	-	Strainer pot	4.7.2.9	D13211E4528
X	-	-	-	Acid pot	4.7.2.9	D13211E4534
X	-	-	-	Exhaust gas cooler	4.7.2.9	D13211E4556
X	-	-	-	Lube oil cooler	4.7.2.9	D13211E4567
X	-	-	-	Vent condenser	4.7.2.9	D13211E4581
X	-	-	-	Engine water cooler	4.7.2.9	D13211E4582
X	-	-	-	Heat exchanger	4.7.2.9	D13211E7750
X	-	-	-	Bottom section	4.7.2.9	D13211E4508
X	-	-	-	Top shell	4.7.2.9	D13211E4526
X	-	-	-	Evaporator tube shell	4.7.2.9	D13211E4527
X	-	-	-	Tank assembly, fuel	4.7.2.10	D13211E7758
X	-	-	X	Pump, distillate	4.7.2.11	D13211E7767
X	-	-	X	Pump, feed	4.7.2.11	D13211E7770
-	X	-	-	Performance, 500 hour	4.7.2.12	3.11
-	-	-	X	Performance, 8 hour	4.7.2.13	3.11
X	X	-	X	Engine performance	4.7.2.14	3.12

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TABLE III. Test schedule. - Continued

Individual component	Preproduction	Initial production	Quality conformance	Test	Test paragraph	Requirement paragraph
1	2	3	4	5	6	7
-	-	X	-	Mobility	4.7.2.15	3.13
-	X	X	-	Electromagnetic interference	4.7.2.16	3.14
-	-	X	-	Transportability	4.7.2.17	3.23

4.7.2.3 Environmental.

4.7.2.3.1 High temperature. Distillation units shall be tested in accordance with MIL-STD-810, Method 501, Procedure II. The maximum test temperature for storage shall be plus 160° F (plus 71° C) and for operating shall be plus 110° F (plus 43° C). Nonconformance to 3.6 shall constitute failure of this test.

4.7.2.3.2 Low temperature. Test shall be conducted in accordance with MIL-STD-810, Method 502, Procedure I. The minimum test temperatures for storage shall be minus 30° F (minus 34° C), and the minimum operating temperatures shall be minus 25° F (minus 32° C). Nonconformance to 3.6 shall constitute failure of this test.

4.7.2.4 Safety. Safety evaluation shall be accomplished throughout testing. Nonconformance to 3.7 shall constitute failure of this test.

4.7.2.5 Human factors. The distillation units shall be evaluated throughout testing to determine the human factor characteristics. Nonconformance to 3.8 shall constitute failure of this test.

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4.7.2.6 Noise level measurement.

4.7.2.6.1 Operator's station. Measure the sound level at the operator's station in accordance with MIL-STD-1474 for steady-state noise while the unit is operating at rated capacity. Nonconformance to 3.8.1 shall constitute failure of this test.

4.7.2.6.2 Hazard precautions. Measure the noise level at all personnel-occupied areas, such as the operator's station, maintenance and control positions, and passenger areas, while operating at the maximum noise condition for those positions. Should the noise level exceed 85 dB(A) at any position, failure to provide the precautions specified in 3.8.1 shall constitute failure of this test.

4.7.2.7 Reliability. Using the MTBF specified in 3.9 the initial production distillation unit shall be tested as specified herein, with "ACCEPT" or "REJECT" criteria in accordance with MIL-STD-781, Test Plan II. Continue testing until either an "ACCEPT" or "REJECT" decision is reached. Failure, for the purpose of this test, is defined as any of the following malfunctions and which the operators cannot remedy by adjustment, repair, or replacement action within 30 minutes (excluding waiting or startup time) utilizing the resources available to the parent unit or organization when deployed:

(a) Initial distillate is not produced within 1 hour and 15 minutes after start of engine.
(b) Average distillate flow rate is less than 150 gph between each acid cleaning period.
(c) Average water-to-fuel ratio (weight basis) is less than 150 to 1 during each 24-hour period.
(d) Temperature of distillate is more than 15° F (8.5° C) above the temperature of feed water.
(e) The total chloride content of the distillate exceeds 20 mg/L in any 8-hour period.
(f) Any failure of welded seam, connection, control, valve, instrument, or any other component or part which is essential to the operation that the repair of cannot be delayed until the next schedule maintenance period.
(g) Brine waste (blowdown) flow rate is less than one-third the feed-water flow rate.

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(h) Concentration of the brine waste is more than 3.0 times the salinity of the feed water.
(i) Any malfunction which would create a serious personnel safety hazard.

Simultaneous related malfunctions are considered as one failure.

4.7.2.7.1 Test procedure. The distillation unit shall be set up, operated, and maintained in accordance with the applicable technical manual supplied with the unit. Refer to the technical manual for complete operational instructions in order to simulate actual use. Operate equipment continuously for periods of 20 hours per day at rated capacity, then shut down unit. Daily preventive maintenance and cleaning is allowed during the 4-hour nonoperating period (not to be included in assessing MTBF).

4.7.2.7.2 Failure criteria. Nonconformance to 3.9 shall constitute failure of this test.

4.7.2.8 Maintainability. Maintenance operations required during initial production testing shall be accomplished in accordance with 3.10. Nonconformance to 3.10 shall constitute failure of this test.

4.7.2.9 Hydrostatic pressure. Prior to being assembled and before insulation is applied, each vent condenser, exhaust gas cooler, oil cooler, engine water cooler, heat exchanger, strainer pot assembly, acid pot assembly, top shell, bottom section, and evaporator tube shell, shall be subjected to a hydrostatic pressure without leakage or damage at not less than that specified on the applicable drawing for not less than 30 minutes. For the purpose of this test the top shell and bottom section shall be assembled together. On components having more than one compartment, each compartment shall be tested independently. The test fluid shall contain a certified food coloring. At the end of 30 minutes, the component shall be wiped with a clean white cloth and examined for deformation, cracks, leaks, and broken welds. Discoloration of the cloth shall be considered as evidence of leaks. All components shall be cleaned as specified in 3.24.7 before and after testing. For hydrostatic testing all traces of water shall be removed for exterior surfaces before final leak testing is begun. Evidence of deformation, cracks, leaks, and broken welds shall constitute failure of this test.

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4.7.2.10 Pneumatic pressure. The gasoline fuel tank shall be subject to a pneumatic pressure without leakage or damage at not less than that specified on the applicable drawing while immersed in water for not less than 15 minutes. Evidence of escaping air shall constitute failure of this test.

4.7.2.11 Pumps. Pumps shall be tested in accordance with the Centrifugal Pump Standards, Rating Standards, and Test Code Standards of the Hydraulic Institute. Inability of the feed and distillate water pumps to operate at capacities specified on the applicable drawing shall constitute failure of this test.

4.7.2.12 Performance for 500 hours.

4.7.2.12.1 Test equipment. Test equipment shall include but not be limited to the following:

Suction hose	- 1 inch inside diameter (ID) - 11-1/2 NPSH couplings.
Discharge hose	- 1 inch ID - 11-1/2 NPSH couplings.
Discharge hose	- 1-1/2 inch ID - 11-1/2 NPSH couplings.
Citric acid	- Monohydrate, technical, crystalline form, MIL-C-11029.
Gasoline	- MIL-G-3056, Type I or MIL-G-5572, grade as applicable.
Scale, weighing	- Accurate to within $\pm$2 percent at any scale reading.
Pail	- 3-gallon capacity, graduated in quarts and gallons.
Fuel oil, diesel	- MIL-F-16884 (F-75) or VV-F-800, grade as applicable.

4.7.2.12.2 Procedure. Distillation unit shall be operated continuously for not less than 500 hours in accordance with Appendix I. The feed water shall be seawater that has been strained through a strainer conforming to MIL-S-12165, Type II. Seawater shall be natural, having the following characteristics:

Total chlorides, mg/L as chlorides	17,000 to 20,000
Bicarbonate alkalinity, mg/L as $CaCO_3$	105 to 125
Magnesium, mg/L as Mg	1,100 to 1,400
Calcium, mg/L as Ca	350 to 450

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Sulphate, mg/L as SO_4	2,500 to 2,800
pH	7.2 to 8.5
Temperature	plus 32° F (0° C) to plus 90° F (plus 32° C)

Shutdown periods shall not exceed 2 hours per 24 hours for maintenance purposes, except for schedule maintenance as specified in the Appendix.

4.7.2.12.3 Test data. The following test data shall be recorded beginning 2 hours after start of engine, and furnished to the contracting officer:

(a) Temperature of feed water and distillate not less than once every hour.
(b) The gallons of distillate produced each hour as determined by collecting and weighing the distillate produced for 30 minutes once each hour.
(c) Weight of fuel consumed in 2 hours not less than once every 8 hours.
(d) Water-to-fuel ratio using calculated weight of distillate produced and fuel consumed in 1 hour as determined using recorded weight of distillate and fuel.
(e) Amount of total chlorides in distillate not less than once in each 8-hour period as determined in accordance with APHA Standard Methods for Examination of Water and Wastewater.
(f) The following characteristics of feed water as determined in accordance with APHA Standard Methods for Examination of Water and Wastewater not less than once in each 72-hour period:

Total chlorides, mg/L as chlorides
Bicarbonate alkalinity, mg/L as $CaCO_3$
Magnesium, mg/L as Mg
Calcium, mg/L as Ca
Sulfate, mg/L as SO_4
pH

(g) Flow rate of brine water (blowdown) not less than once each 8 hours as determined by subtracting the distillate flow rate from the feed-water flow rate.

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(h) Amount of total chlorides in brine water as determined not less than once in each 24-hour period in accordance with APHA Standard Methods for Examination of Water and Wastewater.
(i) Lube oil temperature in engine oil sump and gallery.

4.7.2.12.4 Failure criteria. Evidence of any one of the following conditions shall constitute failure of this test:

(a) Initial distillate is not produced within 1 hour and 15 minutes after start of engine.
(b) Average distillate flow rate is less than 150 gph between each acid cleaning period.
(c) Average water-to-fuel ratio (weight basis) is less than 150 to 1 during each 24-hour period.
(d) Temperature of distillate is more than 15° F (8.5° C) above temperature of feed water.
(e) The total chloride content of the distillate exceeds 20 mg/L in any 8-hour period.
(f) Leakage at any welded seam or connection.
(g) Any control, valve, or instrument fails to operate properly.
(h) Mechanical failure of any component or part.
(i) Shutdown periods for maintenance exceeds 2 hours per 24 hours.
(j) Brine waste (blowdown) flow rate is less than one-third the feed-water flow rate.
(k) Concentration of the brine waste is more than 3.0 times the salinity of the feed water.
(l) Engine lube oil temperature in excess of limits specified in 3.12.1 and 3.12.2, as applicable.

4.7.2.13 Performance for 8 hours. Each production distillation unit shall be operated continuously for not less than 8 hours in accordance with the Appendix. The feed water shall be fresh water conforming to EPA National Interim Primary Drinking Water Regulations except that the combined calcium and magnesium hardness shall not exceed 20 mg/L expressed as $CaCO_3$. In order to determine internal leaks, sodium chloride shall be added to the feed water in the amount necessary to provide a total chloride content of not less than 17,000 nor more than 20,000 mg/L expressed as chlorides. Hardness and chlorides shall be

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determined in accordance with APHA Standard Methods for Examination of Water and Wastewater. Temperature of the feed water shall be between plus 32° F and plus 90° F (0° C and plus 32° C). Test equipment shall be as specified in 4.7.2.11.1.

4.7.2.13.1 Test data. The following test data shall be recorded beginning 2 hours after start of engine and a report of the tests shall be furnished to the contracting officer:

(a) Temperature of feed water and distillate not less than once every hour.
(b) The gallons of distillate produced each hour as determined by collecting and weighing the distillate produced for 30 minutes once each hour.
(c) Weight of fuel consumed in 2 hours.
(d) Amount of total chlorides in distillate as determined in accordance with APHA Standard Methods for Examination of Water and Wastewater. Disapproval of the report shall constitute failure of this test.

4.7.2.14 Engine performance (gasoline and diesel). Prior to examination and tests of the distillation unit, the engine which will be installed in the first article of initial production units shall be tested in accordance with Test Method 2000 of MIL-STD-1400, except that Test Method 2400 shall not apply. One engine out of each lot of 50 or less shall be tested in accordance with Test Method 3000, MIL-STD-1400. Each engine furnished, except the engines selected for the preproduction model, initial production test, and production control tests (Test Method 3000), shall be tested in accordance with Test Method Series 4000, MIL-STD-1400, except that the engine manufacturer's standard commercial run-in may be used in lieu of Test Method Series 4000 provided it equals or exceeds the requirements of Test Method Series 4000, MIL-STD-1400. Failure to meet any requirements in the above tests shall constitute failure of this test.

4.7.2.15 Mobility. The distillation unit, mounted on the specified trailer shall be towed with a Government-loaned truck as follows:

(a) Tow the distillation unit for a distance of 300 miles ±1 mile, over smooth, hard-surfaced primary highways at variable speeds up to 50 mph.

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(b) Tow the distillation unit for a distance of 150 miles ±1 mile, over unimproved secondary roads at variable speeds up to 25 mph.
(c) Tow the distillation unit for a distance of 50 miles ±1 mile, over open, rolling, and hilly cross-country terrain at variable speeds up to 10 mph.

4.7.2.15 Failure criteria. Nonconformance to 3.13 shall constitute failure of this test.

4.7.2.16 Electromagnetic interference. The preproduction model shall be tested to determine conformance to 3.14. The contractor shall furnish the contracting officer a report of tests from an independent tester as required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.7.2.17 Railroad car impact. The preproduction uncrated distillation unit shall be subjected to the guided-impact railroad-car test as specified for guided-impact tests in accordance with MIL-STD-1186, Appendix A as modified. Antiskid plates shall not be used for this test. The railroad car shall be humped at speeds of 4, 6, and 8 mph respectively against 5 empty freight cars with their brakes set and draft gear extended. Repeat the procedure at not less than 8 mph with the rear of the distillation unit in the direction of the 5 cars (total of 4 impacts). After each impact the distillation unit shall be examined for damage such as shifting of components, loosening or breaking of welds, holddowns, ties, stays, or any visible damage attributable to the impact test. Evidence of any of the above damage shall constitute failure of this test.

4.8 Inspection comparison. The Government may select distillation units any time during the contract production period and subject these distillation units to the examination marked "X" in Column 2 of Table II and the tests marked "X" in Column 3 of Table III to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on distillation units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production distillation units. In addition to any test specified as part of the inspection comparison,

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the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.8.1 Inspection failure. Failure of an inspection comparison distillation unit to meet any requirement specified herein during and as a result of the examination and tests specified in 4.8 shall be cause for rejection of the inspection comparison distillation unit and shall be cause for refusal by the Government to continue acceptance of production distillation units until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on distillation units previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all distillation units accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.9 Inspection of packaging. The preservation, packing, and marking of each distillation unit for Level A or Level B shall be examined to determine conformance to MIL-D-3209, and in addition when capped units are required, examination shall include conformance to Figures 2, 3, and 4 of this specification. Commercial preservation, packing, and marking shall be examined for conformance to MIL-STD-1188.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be Level A, Level B, or Commercial, as specified (see 6.2). Level A or Level B preservation, packing, and marking shall be in accordance with MIL-D-3209 for mobile shipment. Commercial preservation, packing, and marking shall be in accordance with MIL-STD-1188, except a shipping container is not required. When Level A packing is specified, each distillation unit shall be protected with a plywood cap in accordance with Figures 2, 3, and 4 of this specification.

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6. NOTES

6.1 Intended use. The military design thermocompression distillation units are intended to supply 150 gph of potable water from natural seawater and fresh water sources. This distillation unit is a component of Distillation Equipment Set: Water Thermocompression Type, 150 GPH; Department of the Army Supply Catalog SC 4620-97-CL-E04.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of distillation unit required (see 1.2).
(c) Time frame required for submission of preproduction model (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examinations and tests. When the Government will conduct some but not all of the preproduction examinations and tests, the contracting officer should specify which examinations and tests will be conducted by the Government and which examinations and tests shall be conducted by the contractor (see 3.2).
(e) When initial production inspection is required, and number of distillation units to be furnished (see 3.3).
(f) When Government-furnished property is required (see 3.21).
(g) When Government-loaned property is required (see 3.22).
(h) When railway hump test is not required (see 3.23.1).
(i) When the welding process is not at the contractor's option (see 3.24.3.1).
(j) When ALCLAD aluminum may be ground, filed, chipped, or hammered (see 3.24.3.3.1).
(k) Degree of preservation, packing, and marking required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production distillation units from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish distillation units conforming to this specification.

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6.4 Health hazards. For Army procurement, the Surgeon General of the Army will act as advisor to make the necessary determination on health hazards.

6.5 Definitions.

6.5.1 Supplementary quality assurance provisions (SQAP). A SQAP is a contractual requirement that supplements Section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.5.2 End Item Final Inspection Requirement (EIFIR). An EIFIR is a check list of quality characteristics prepared by the Government which the contractor must follow during his final inspection of the completed end item to verify complete and functional conformance to contract requirements prior to submission for final acceptance.

6.6 Government-furnished property. The contracting officer should arrange to furnish the property specified in 3.21, when applicable (see 6.2).

6.7 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.22, when applicable (see 6.2).

6.8 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each distillation unit.

6.9 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of distillation units.

6.10 Noise limits. When the noise level of the distillation unit exceeds 85 dB(A), the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 (MANUALS Paragraph) shall include the requirement for hearing protection, the noise level of the distillation unit, the distance from the distillation unit at which 85 dB(A) will always be met, and what operational situation or system configuration will reduce the level to 85 dB(A).

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6.11 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the distillation unit covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.12 Supplementary quality assurance provisions (SQAP) records. The contracting officer should require the contractor to maintain records of all SQAP inspections. A suggested paragraph is as follows:

> "The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.13 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.4).

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activities:
Army - MD
DLA - CS

Project 4620-A029

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APPENDIX

OPERATING PROCEDURE FOR DISTILLATION UNIT TESTING

10. SCOPE

10.1 Scope. This appendix explains operating procedures to be used in testing a distillation unit to determine conformance to the requirements of this specification. Location of various components identified by nomenclature or number are as shown in Figure 1A for Type I and Figure 1B for Type II, as applicable.

20. PROCEDURES

20.1 Preliminary. The following procedures shall be accomplished before the distillation unit is started:

(a) Level the unit.
(b) Check all hose couplings for washers.
(c) Connect hose from unit strainer to feed-water supply, with hose strainer.
(d) Remove caps from brine overflow and distillate outlets of heat exchanger and connect respective hoses.
(e) Check oil level in engine.
(f) Check oil level in the gearcase of the compressor.
(g) Check all pressure fittings and lubricate as necessary.
(h) Close all drain cocks and drain valves.
(i) Fill the fuel tank or connect to external fuel supply.
(j) Check engine water cooler. Gage glass should indicate cooler to be not less than 3/4 full. Add clean distillate or fresh water as necessary through valve BV 13.
(k) Check battery terminals and electrolyte level.
(l) Check all belts.

20.2 Type I distillation unit. The following starting procedures shall be used:

(a) Close valves BV 5 and BV 13 on the engine water cooler and all drain valves of distillation unit.
(b) Open valves 1, 2, 3, 4, 6, 10, 12, 21, 22, and 7.
(c) Open valve DV 17, drain compressor of all condensate. Close valve DV 17 when compressor is drained.

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(d) Turn the compressor over by hand. This shall be done every time unit is started.
(e) Start engine in accordance with the engine manufacturer's instructions.
(f) After about 3 minutes, engage engine clutch to start compressor.
(g) Adjust valve CV 1 to obtain feed-water flow rate of 90 gph (1-1/2 gallons per minute (gpm)) and allow feed water to flow for 25 minutes, then close valve CV 1; or open valve CV 1 and allow feed water to flow until water reaches level of upper end of evaporator tubes as viewed through sight glass, then close valve CV 1.
(h) Close valve DV 7 until a running drip is maintained.
(i) Close valve CV 4 and then reopen approximately 1-1/2 turns.
(j) Slowly close the evaporator bypass valve BV 3 but do not exceed 5 pounds per square inch gage (psig) compressor discharge gage PG-1.
(k) When the bypass valve BV 3 is closed (in approximately 1 hour) adjust the feed-water flow rate valve CV 1 to maintain between 20 and 30 inches of water column on the compressor suction gage PG-2.
(l) Adjust valve CV 4 to show a wisp of steam.
(m) For Type II distillation unit, close backpressure valve BPV 31 and adjust feed water valve CV 1 accordingly.

20.3 Type II distillation unit. The starting procedures for Type II distillation units are precisely as described in Paragraph 20.2, except that the backpressure valve BPV 31 as shown in Figure 1B is open in operation 20.2(b).

20.4 Operation. Upon completion of starting procedure, the distillation unit shall be operated as follows:

(a) Adjust the feed-water flow rate until the pressure is indicated on the compressor suction gage PG-2 reads between 20 and 30 inches of water.

CAUTION

The feed-water flow rate must be adjusted in small increments with several minutes lapse between each increase as it takes time for each adjustment to show the effect.

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(b) If the compressor suction gage PG-2 begins to go to the vacuum side, decrease the feed-water flow rate 30 gph (1/2 gpm) until the pressure returns to the positive side. If after several minutes the gage still indicates a vacuum again decrease the feed-water flow rate 30 gph (1/2 gpm). Repeat this procedure until the pressure returns to the positive side.

(c) If compressor suction gage PG-2 continues to go to the vacuum side, reopen valve BV 3 and partially close valve CV 1 and wait until gage PG-2 registers a positive pressure. Repeat previous procedure as in 20.2(j) and 20.2(k) until valve BV 3 is closed.

(d) When the suction pressure gage PG-2 begins to show an increase in pressure, increase the feed-water flow rate until suction pressure gage PG-2 reads between 20 and 30 inches of water.

(e) When the unit has reached a normal operating condition, make periodic inspections to ascertain that the equipment is operating efficiently.

20.5 Acid cleaning. At the end of each 8 hours of operation, the unit shall be acid-cleaned, while in operation, using the following procedures:

(a) Close acid pot drain valve DV 19.

(b) Loosen the two thumb screws from the top of the acid pot and remove cover.

(c) Add citric acid crystals to the level indicated by the mark on the inside of the pot (1-1/2 pounds of citric acid).

(d) Replace cover and tighten thumb screws.

(e) Open valves BV 15 and BV 14.

(f) Close valve BV 2.

(g) Allow the feed water to flow through the acid pot for 5 minutes, and then reduce feed-water flow rate, valve CV 1, to reduce tendency of the distillation unit to go on vacuum.

(h) Open valve BV 2 and then close valves BV 14 and BV 15 to return distillation unit to normal operation.

(i) Open acid pot drain valve DV 19 and leave open until next acid cleaning.

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30. MAINTENANCE

30.1 Scheduled maintenance. Between each 200 and 250 hours of operation, the following maintenance shall be accomplished:

(a) While the distillation unit is operating, dissolve approximately 5 pounds of citric acid in distilled water using the pail. Place the feed-water intake hose, the brine water discharge hose, and the distilled water discharge hose into the pail and continue operation for 5 minutes. Then disengage clutch to stop compressor, and open evaporator drain valve DV 16 and compressor drain valve DV 17. When evaporator is empty, refill the feed water and drain evaporator again. Shut off engine.
(b) When distillation unit is cooled, remove end plates on exhaust gas cooler and clean carbon from exhaust gas cooler tubes. Reassemble exhaust gas cooler.
(c) Open drain cocks on engines and drain all water from engine water cooler. When engine water cooler and engine are completely drained, close drain cocks and refill with distilled water.
(d) Lubricate and service the distillation unit in accordance with preventive maintenance schedule.

40. UNIT SHUTDOWN

40.1 Distillation unit. The following functions are to be performed in sequence when shutting down the distillation unit:

(a) Fill engine water cooler with distillate by opening valve BV 5 and closing valve BV 6.
(b) When water level reaches 3/4 height of gage glass, open valve BV 6 and close valve BV 5.
(c) Open valves BV 3 and CV 4.
(d) Disengage clutch and slow engine to idle speed.
(e) Fill evaporator to operating level with feed water.
(f) Close valve CV 1.
(g) Open valves DV 17 and CV 7.
(h) Push magneto switch control in to stop engine.

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(i) When the distillation unit is to be permanently shut down, acid-clean the unit as specified in 30.1(a) except that distilled or tap water shall be used in lieu of seawater. After completion of acid cleaning, flush all components with distilled or tap water, open all valves, and ascertain that all components, hoses, gages, etc., are completely drained.

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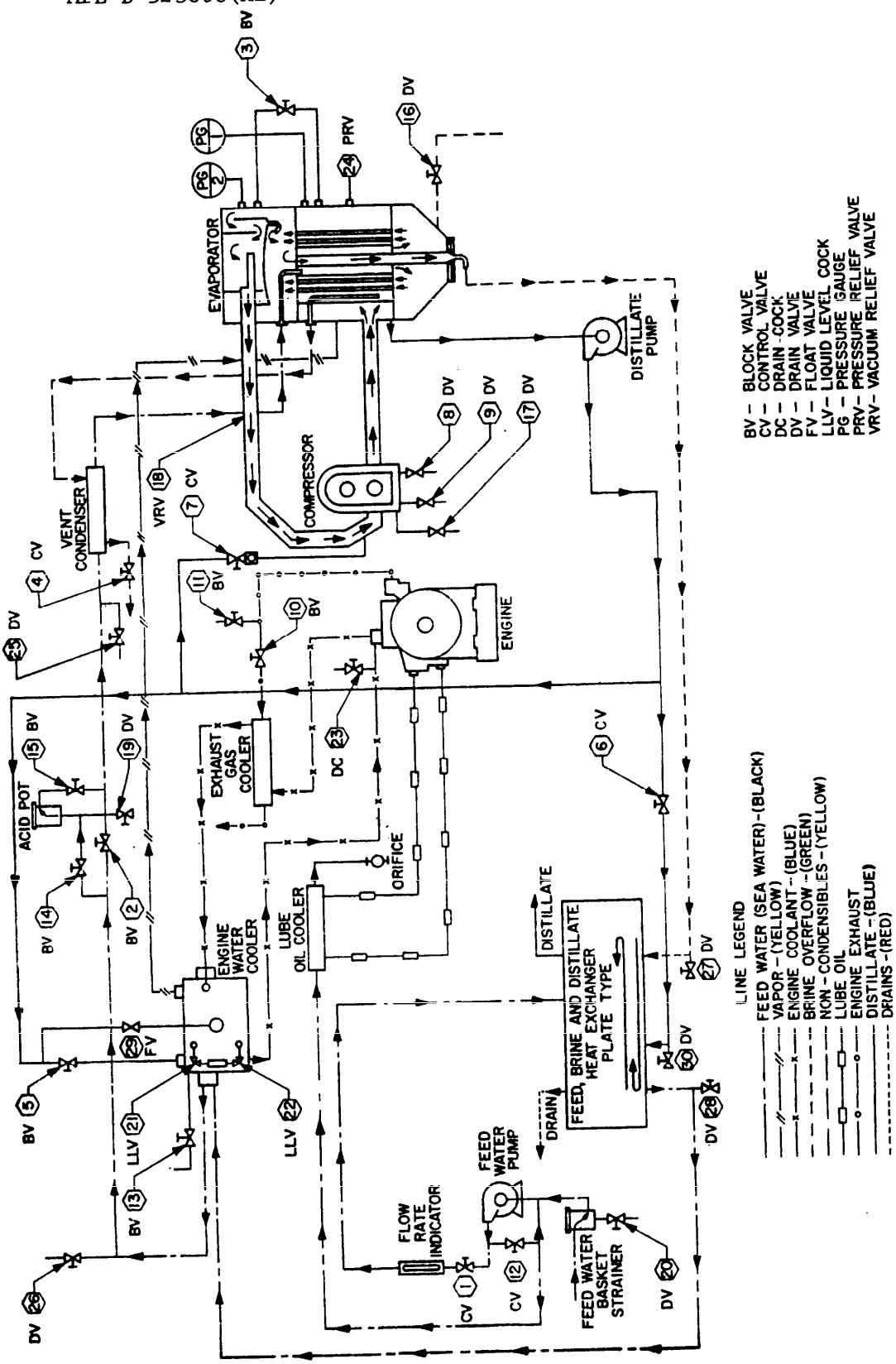


FIGURE IA. Flow diagram, Type I.

X-135B

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X2133

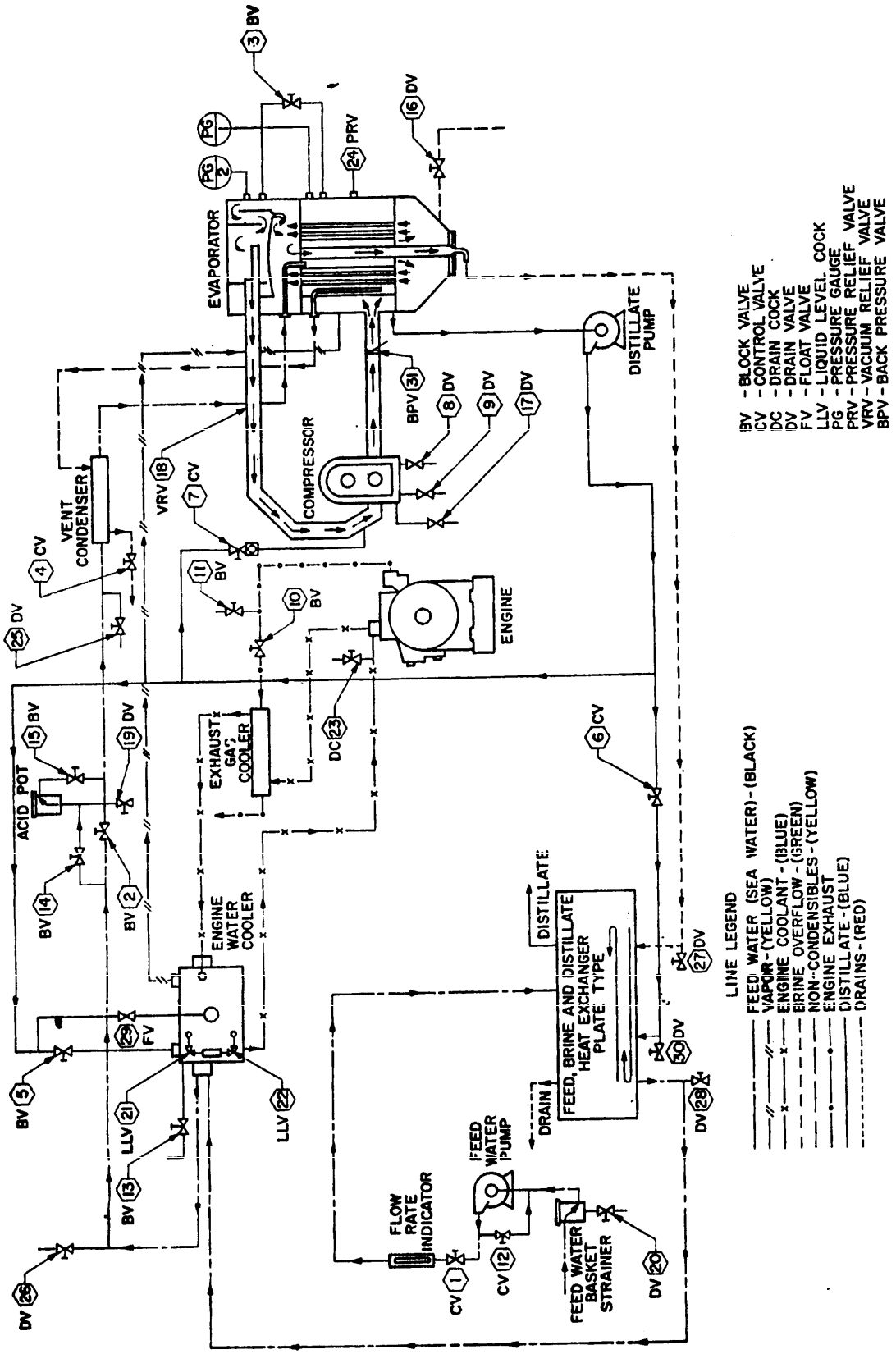


FIGURE IB. Flow diagram, Type II.

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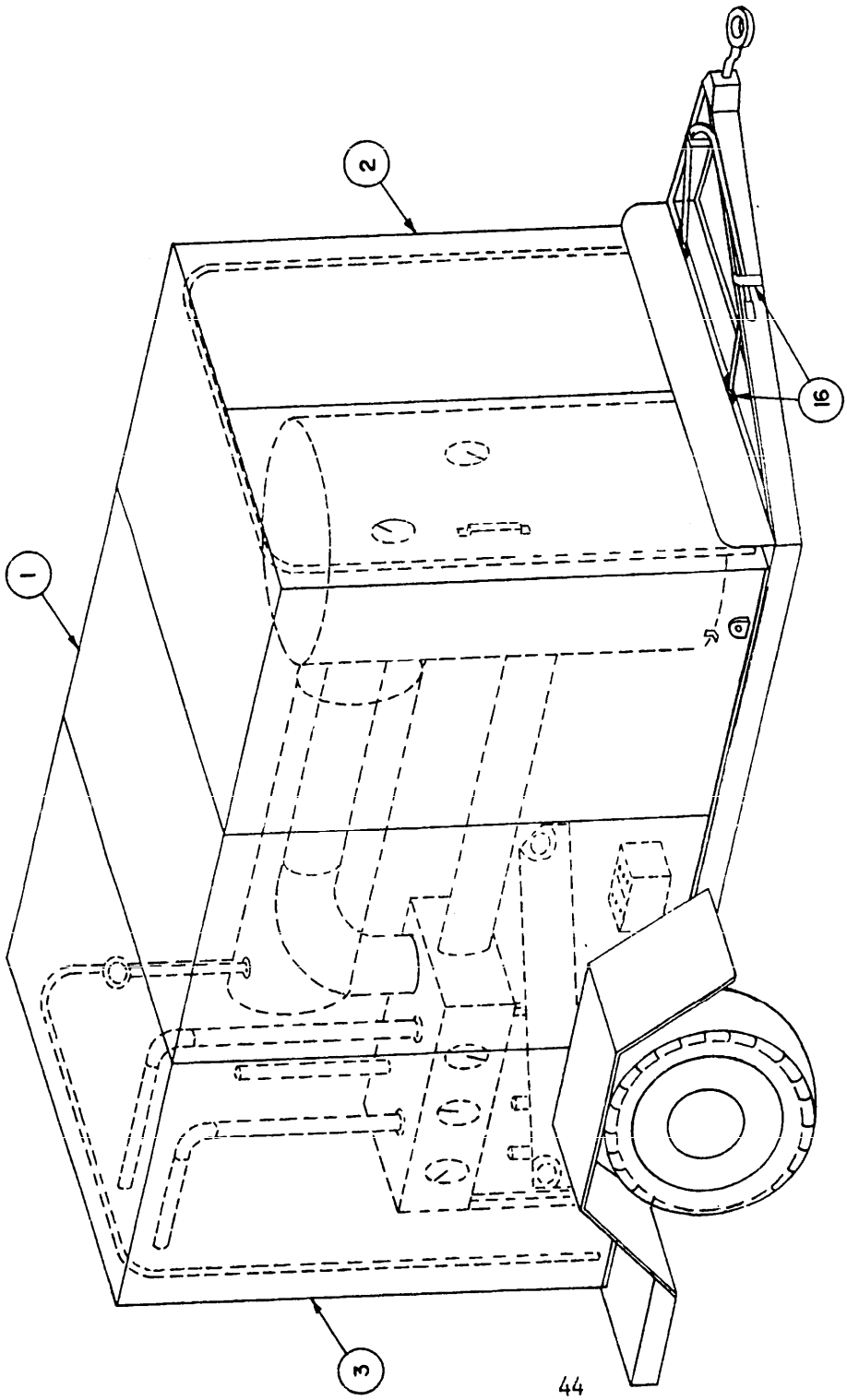
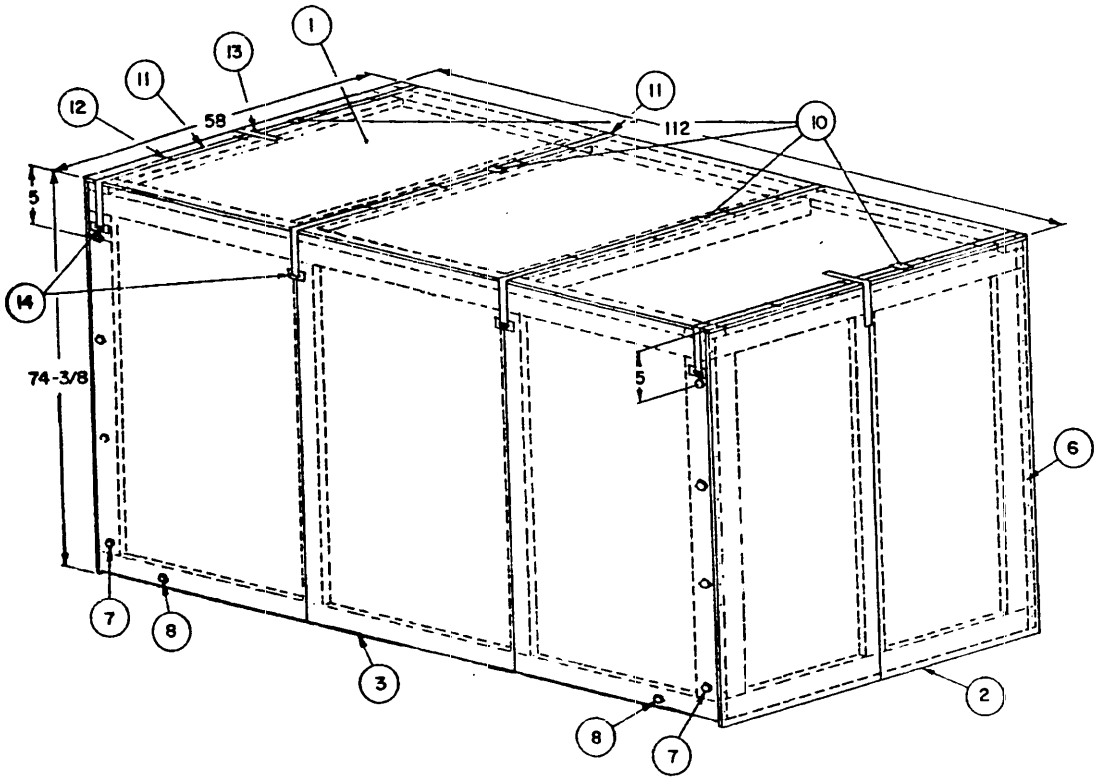


FIGURE 2. Packing Level A: capped distillation unit. X1856

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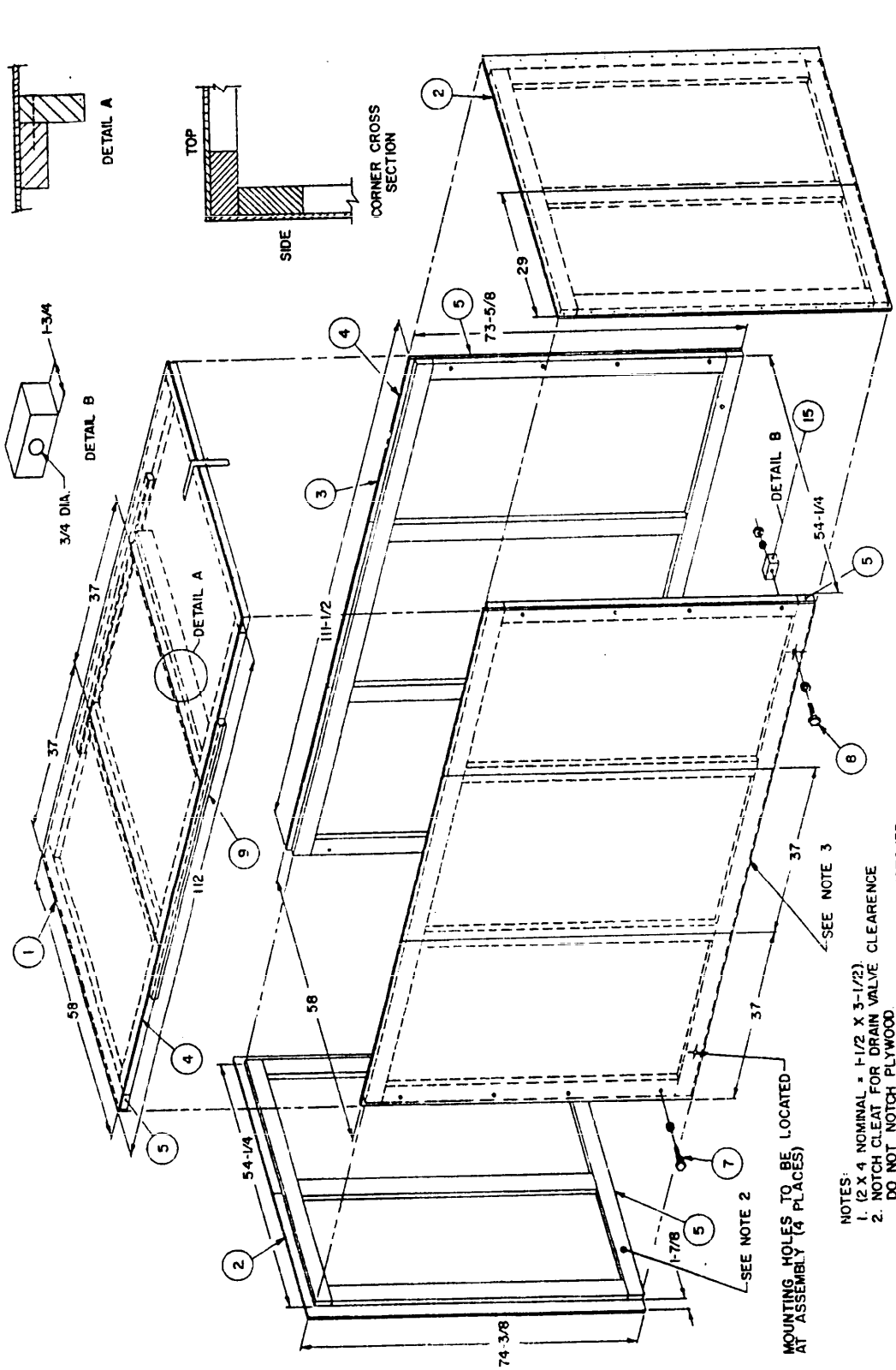
FIND NO.	QTY	NOMENCLATURE
1	1	TOP PANEL 112 X 58
2	2	END PANEL 72-3/4 X 58
3	2	SIDE PANEL 111-1/2 X 72
4	AR	ALL PLYWOOD PANELS
		(NN-P-530) GROUP A(PS-51, TYPE
		I, GRADE 3-4) OR GROUP B (PS-1,
		GRADE C-D INTERIOR W/ EXTERIOR
		GLUE), 1/4 -IN THICKNESS
5	AR	ALL TOP, SIDE, AND END CLEATS
		NOM. 2 X 4
6	AR	6d (MIN.) CEMENT COATED COOLERS
		OR SINKERS (FF-N-105) TYPE II
		STYLE 7 OR 8
7	16	3/8 LAG SCREW WITH WASHERS 3-1/2 LG
8	4	5/8 BOLTS 6" LG WITH HEX NUTS
		FLAT WASHERS, AND LOCK WASHERS
9	2	NOM. 2 X 2 X 60
10	4	STRAPPING SEALS
11	AR	STRAPPING QQ-S-781 TYPE I OR IV,
		FINISH B 1.25 X .035
12	AR	STAPLES PPP-B-601 PARA. 30.4
13	AR	STRAPPING QQ-S-781 TYPE II
		FINISH B
14	8	ANCHOR PLATES
15	4	SPACER, 2 X 2 X 1-3/4 WIDE
16	AR	PRESSURE SENSITIVE TAPE (PPP-T-60)
		TYPE IV

NOTE:
DIMENSIONS, AS GIVEN, ARE TYPICAL BUT MINOR MODIFICATIONS MAY BE REQUIRED TO CONFORM TO TRAILER SIZE AS FURNISHED.

FIGURE 3. Packing Level A: cap assembly

X1857B

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X1858A

FIGURE 4. Cap assembly details.

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